

ZEROS OF HIGH DERIVATIVES OF SPARSE FOCK SERIES

ZHIJIE HE

ABSTRACT. For $4/3 \leq p < 2$, we prove that the explicit entire function $F_p(z) = \sum_{j \geq 1} z^{\lfloor j^p \rfloor} / \sqrt{\lfloor j^p \rfloor!}$ has order two, type $1/2$, and a zero of every sufficiently high derivative in each nonempty open set. For $p > 4/3$, a discrete concavity argument identifies all derivative zeros on fixed annuli with small perturbations of explicit circular grids. It gives a sharp covering scale, eventual simplicity away from the origin, and a limiting zero measure. At $p = 4/3$, a compactness argument for bilateral Gaussian series gives the sharp annular covering scale $n^{-1/4}$ without discarding the phases introduced by integer parts. These are deterministic constructions for the cofinite formulation of Erdős Problem 906; no first-solution claim is made.

1. STATEMENTS AND CONTEXT

For a nonzero entire function g , let $\mathcal{Z}(g)$ be its zero set. The property under consideration is

$$\forall U \subset \mathbb{C} \text{ nonempty open, } \exists N(U) \quad \forall n \geq N(U) : \quad \mathcal{Z}(f^{(n)}) \cap U \neq \emptyset. \quad (1.1)$$

It is equivalent to density of $\bigcup_k \mathcal{Z}(f^{(n_k)})$ for every strictly increasing infinite sequence (n_k) of nonnegative integers. Indeed, an infinite set of exceptional orders for one open set gives a sequence violating density; conversely, an increasing infinite sequence eventually exceeds any fixed threshold. We require f to be transcendental, since polynomials make the question trivial.

This is the question in Erdős [2, p. 72, IV.1(i)], now numbered 906. The adjacent interpolation question, associated with Problem 229, concerns prescribed discrete sets and derivative orders selected during the construction. Barth and Schneider [1, Theorem 1] settle that interpolation problem. It does not itself provide the cofinite quantifier in (1.1). MacLane's derivative universality theorem [5] is another related density result.

Hou [4, Theorem 1.2] proves (1.1) by a bounded random Fock series, with a finite quadratic exponential growth bound. His public repository accompanies the paper with a Lean formalization; we have not rerun it. His introduction also records earlier probabilistic proposals by Almeida and Chojecki. Our purpose is an explicit sparse series and its local zero geometry. Neither the existence assertion nor order-two finite-type existence is claimed as a first result here.

Fix $p \in [4/3, 2)$ and set

$$\nu_j = \lfloor j^p \rfloor, \quad q_j = \nu_{j+1} - \nu_j, \quad \beta = 1 - \frac{1}{p}, \quad F_p(z) = \sum_{j=1}^{\infty} \frac{z^{\nu_j}}{\sqrt{\nu_j!}}. \quad (1.2)$$

For $0 < a < b$, put $A_{a,b} = \{z : a \leq |z| \leq b\}$ and $h_n(E) = \sup_{w \in E} \text{dist}(w, \mathcal{Z}(F_p^{(n)}))$. All constants below may depend on fixed radii and on p , but not on n or on points in the indicated sets.

Theorem 1.1. *The function F_p is transcendental entire, satisfies (1.1), and has*

$$\log M_{F_p}(r) = \frac{1}{2}r^2 + O_p(\log r) \quad (r \rightarrow \infty). \quad (1.3)$$

In particular, its order is exactly two and its type is $1/2$.

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(i) If $4/3 < p < 2$, then for every $R > 0$,

$$h_n(\{|z| \leq R\}) \leq C_{R,p}(n^{\beta-1/2} + n^{-\beta}) \quad (1.4)$$

for all sufficiently large n . On every fixed annulus $A_{a,b}$, this implies, and can be sharpened to,

$$c_{a,b,p}n^{\beta-1/2} \leq h_n(A_{a,b}) \leq C_{a,b,p}n^{\beta-1/2}. \quad (1.5)$$

(ii) If $p = 4/3$, then

$$c_{a,b}n^{-1/4} \leq h_n(A_{a,b}) \leq C_{a,b}n^{-1/4} \quad (1.6)$$

for all sufficiently large n .

The upper-bound proof in (1.6) is a compactness argument; it does not give an explicit numerical value for $C_{a,b}$.

For $\nu_j \geq n$, write

$$m_j = \nu_j - n, \quad A_j = \frac{\sqrt{\nu_j!}}{m_j!}, \quad T_j(z) = A_j z^{m_j}, \quad \rho_j = \left(\frac{A_j}{A_{j+1}} \right)^{1/q_j}. \quad (1.7)$$

Dependence on n is suppressed in these four quantities. Define the model centers

$$z_{j,\ell} = \rho_j \exp\left(\frac{(2\ell+1)\pi i}{q_j}\right), \quad 0 \leq \ell < q_j. \quad (1.8)$$

Theorem 1.2. Suppose $4/3 < p < 2$. Fix $0 < a < b$. There are $c > 0$ and a threshold such that, with $\eta_n = \exp(-cn^{2\beta-1/2})$, the disks

$$D\left(z_{j,\ell}, \frac{\rho_j \eta_n}{q_j}\right), \quad a/2 \leq \rho_j \leq 2b, \quad (1.9)$$

are pairwise disjoint and each contains exactly one zero of $F_p^{(n)}$, counted with multiplicity. Every zero in $A_{a,b}$ belongs to one of these disks. Thus every zero on $A_{a,b}$ is eventually simple.

Let $S = \{re^{i\theta} : a < r < b, \alpha < \theta < \alpha + \omega\}$, where $0 < \omega \leq 2\pi$, with angles interpreted on the circle. If $N_n(S)$ counts zeros with multiplicity, then

$$N_n(S) = \frac{\omega}{2\pi}(b-a)\sqrt{n} + O_{a,b,p}(n^\beta + n^{1/2-\beta}). \quad (1.10)$$

Consequently, on $\mathbb{C} \setminus \{0\}$, in the sense of integration against continuous compactly supported functions,

$$\frac{1}{\sqrt{n}} \sum_{F_p^{(n)}(\zeta)=0} \delta_\zeta \longrightarrow \frac{1}{2\pi|z|} dA(z). \quad (1.11)$$

Zeros in the sum are repeated according to multiplicity.

The support in (1.2) is not a Hadamard gap sequence: $\nu_{j+1}/\nu_j \rightarrow 1$. The relevant scale is instead the gap relative to the differentiated saddle width. For $p > 4/3$ it is $n^{1/4} \ll q_j \ll n^{1/2}$ near $\nu_j = n + O(\sqrt{n})$. Gethner's work on final sets [3] and Offord's work on lacunary entire functions [6] provide earlier context. A final-set assertion formulated using infinitely many orders does not alone imply (1.1). Offord's publisher abstract concerns angular zero distribution and growth away from small zero neighborhoods; a complete comparison with his full hypotheses remains outstanding. We make no priority claim for the sparse construction.

2. ANALYTIC PRELIMINARIES AND THE CROSSING RADII

Lemma 2.1. The integers ν_j strictly increase and

$$q_j = pj^{p-1}(1+o(1)) = p\nu_j^\beta(1+o(1)).$$

Writing $Q = n^\beta$, there are constants $\gamma, \Gamma > 0$ such that, for large n ,

$$q_h \geq \gamma Q \quad (\nu_h \geq n), \quad q_h \leq \Gamma Q \quad (n \leq \nu_h \leq n + B\sqrt{n}) \quad (2.1)$$

for each fixed B . Moreover $q_h/(pQ) \rightarrow 1$ uniformly in the latter range. The first active exponent satisfies $m_{j_0} = O(Q)$, where $j_0 = \min\{j : \nu_j \geq n\}$.

Proof. The mean value theorem gives $q_j = p\xi_j^{p-1} + O(1)$ with $j < \xi_j < j+1$, and the floor error has modulus at most one. Since $(j+1)^p - j^p > 1$, the exponents strictly increase. For active indices $j \geq n^{1/p}$, this gives, for example, $\gamma = p/2$ eventually. In a fixed saddle range $j/n^{1/p} \rightarrow 1$ uniformly, giving the upper bound and the uniform equivalent. The exponent immediately preceding ν_{j_0} lies below n , so the same gap estimate bounds m_{j_0} . No monotonicity of the integer gaps is assumed. $\square$

Lemma 2.2. *The series (1.2) is entire, with*

$$F_p^{(n)}(z) = \sum_{j \geq j_0} A_j z^{m_j}. \quad (2.2)$$

It satisfies (1.3).

Proof. For every $\varepsilon > 0$, Cauchy–Schwarz gives

$$\sum_{k \geq 0} \frac{r^k}{\sqrt{k!}} \leq \left(\frac{1+\varepsilon}{\varepsilon} \right)^{1/2} \exp\left(\frac{1+\varepsilon}{2} r^2 \right). \quad (2.3)$$

This proves normal convergence. Cauchy’s integral formula on a slightly larger circle justifies termwise differentiation to every fixed order and gives (2.2). Infinitely many Taylor coefficients are nonzero, so the function is transcendental.

Taking $\varepsilon = r^{-2}$ in (2.3) yields the upper bound in (1.3). For the lower bound choose $k = \nu_j \leq r^2 < \nu_{j+1}$. Then $r^2 - k = O_p(r^{2\beta}) = o(r)$. The elementary integral bounds for the increasing function $\log t$ give

$$k \log k - k + 1 \leq \log(k!) \leq k \log k - k + 1 + \log k. \quad (2.4)$$

Thus

$$\log \frac{r^k}{\sqrt{k!}} = \frac{1}{2} \{k + k \log(r^2/k)\} + O(\log k) = \frac{1}{2} r^2 + O_p(\log r).$$

The last step follows by Taylor expansion in $(r^2 - k)/r^2$, with error $O((r^2 - k)^2/r^2) = o(1)$. Positivity at $z = r$ supplies the required lower bound. Under the definitions $\text{ord } f = \limsup \log \log M_f(r)/\log r$ and $\text{type}_2 f = \limsup \log M_f(r)/r^2$, the order and type follow. $\square$

The term “Fock” refers here to the normalization $1/\sqrt{k!}$; the coefficient sequence of F_p is not square summable, so no membership in the standard Fock Hilbert space is asserted.

For integers $k \geq n$ and $r > 0$ set

$$\Phi_{n,r}(k) = \frac{1}{2} \log(k!) - \log((k-n)!) + (k-n) \log r, \quad d_{n,r}(k) = \Phi_{n,r}(k+1) - \Phi_{n,r}(k).$$

Lemma 2.3 (Discrete curvature). *Writing $m = k - n$, one has*

$$d_{n,r}(k) = \log r + \frac{1}{2} \log(n+m+1) - \log(m+1), \quad (2.5)$$

$$d_{n,r}(k) - d_{n,r}(k+1) = \log \left(1 + \frac{1}{m+1} \right) - \frac{1}{2} \log \left(1 + \frac{1}{n+m+1} \right) > 0. \quad (2.6)$$

For any $B > 0$, uniformly for $0 \leq m \leq B\sqrt{n}$, the difference in (2.6) is at least $c_B/\sqrt{n}$ for all sufficiently large n , where one may take $c_B = 1/(4(B+1))$. If $m/\sqrt{n} \rightarrow r \in [a, b]$, it equals $(1 + o(1))/(r\sqrt{n})$ uniformly.

Proof. The formulas follow by cancellation of factorials. Positivity follows since $n+m+1 \geq m+1$. The inequalities $x/(1+x) \leq \log(1+x) \leq x$ give the lower bound $1/(m+2) - 1/(2(n+m+1))$, which is at least $c_B/\sqrt{n}$ eventually. Expansion of the two logarithms proves the final assertion. $\square$

Lemma 2.4 (Radial grid). *The radii ρ_j strictly increase to infinity for each fixed n , and*

$$\log \rho_j = \frac{1}{q_j} \sum_{s=1}^{q_j} \left\{ \log(m_j + s) - \frac{1}{2} \log(n + m_j + s) \right\}. \quad (2.7)$$

Uniformly for $0 \leq m_j \leq B\sqrt{n}$,

$$\rho_j = \frac{m_j}{\sqrt{n}} + O_{B,p}(Q/\sqrt{n}). \quad (2.8)$$

Conversely, bounded ρ_j imply $m_j = O(\sqrt{n})$. On fixed compact annuli, consecutive crossing radii have spacings

$$c_{a,b,p} \frac{Q}{\sqrt{n}} \leq \rho_{j+1} - \rho_j \leq C_{a,b,p} \frac{Q}{\sqrt{n}}. \quad (2.9)$$

Proof. Equation (2.7) follows from (1.7). Put $g_n(t) = \log(t+1) - \frac{1}{2} \log(n+t+1)$ for $t \geq 0$ and $f_n(t) = \exp(g_n(t))$. Both increase strictly. Formula (2.7) averages g_n over successive disjoint integer blocks, proving strict ordering, divergence, and

$$f_n(m_j) \leq \rho_j \leq f_n(m_j + q_j - 1). \quad (2.10)$$

For $t = O(\sqrt{n})$, $f_n(t) = t/\sqrt{n} + O(n^{-1/2})$, giving (2.8), including $m_j = 0$. The lower bound in (2.10), squared, shows that $\rho_j \leq b$ forces $(m_j + 1)^2 \leq b^2(n + m_j + 1)$, hence $m_j = O_b(\sqrt{n})$.

On the relevant blocks, Lemma 2.3 gives

$$\log \rho_{j+1} - \log \rho_j \geq \frac{c_B(q_j + q_{j+1})}{2\sqrt{n}}. \quad (2.11)$$

Indeed, the mean difference of the indices in the two consecutive blocks is $(q_j + q_{j+1})/2$; apply the curvature bound to every pair and average. On an annulus the blocks also satisfy $m_j \asymp \sqrt{n}$, where the reverse curvature bound is $C/\sqrt{n}$. This proves (2.9). Enlarging the fixed saddle range slightly includes both blocks throughout. $\square$

3. THE TWO-TERM REGIME

Assume throughout this section that $\beta > 1/4$. We give the infinite-tail estimate explicitly.

Lemma 3.1 (Stable tail bound). *Fix $B > 0$ and $L > 0$. There is $c > 0$ such that, with $E_n = cQ^2/\sqrt{n} \rightarrow \infty$, every active j with $m_j \leq B\sqrt{n}$ satisfies*

$$\frac{\sum_{h \neq j, j+1} |T_h(z)|}{|T_j(z)|} \leq (1 + e^L) \frac{e^{-E_n}}{1 - e^{-E_n}} \quad \text{if } |q_j \log(|z|/\rho_j)| \leq L. \quad (3.1)$$

The threshold may depend on B, L, p .

Proof. At $r = \rho_j$, the $q = q_j$ numbers $d_{n,r}(\nu_j), \dots, d_{n,r}(\nu_j + q - 1)$ have sum zero. Each consecutive drop is at least $c_0/\sqrt{n}$, where $c_0 = 1/(4(B+2))$ is valid after increasing the threshold. Thus their first and last members are respectively at least $c_0(q-1)/(2\sqrt{n})$ and at most its negative. For $q \geq 2$, replace this magnitude by $c_0q/(4\sqrt{n})$.

Changing the radius adds $\log(|z|/\rho_j)$ to every full difference. Since $L/q = o(q/\sqrt{n})$, the first and last values retain the corresponding signs with magnitude at least $c_0q/(8\sqrt{n})$. Global monotonicity of $d_{n,r}$ now controls *all* preceding and following blocks, even outside the saddle range. Every active gap is at least γQ by Lemma 2.1; hence each sparse step away from the pair loses at least $E_n = c_0\gamma^2Q^2/(8\sqrt{n})$ in logarithmic modulus.

The left tail is at most $|T_j| \sum_{s \geq 1} e^{-sE_n}$. The right tail is at most $|T_{j+1}| \sum_{s \geq 1} e^{-sE_n}$, and $|T_{j+1}|/|T_j| \leq e^L$. Summing these geometric series proves (3.1); the actual left tail may terminate earlier. $\square$

Proposition 3.2 (Rouché transfer). *Uniformly for $0 \leq m_j \leq B\sqrt{n}$, every disk $D(z_{j,\ell}, \rho_j \eta_n / q_j)$ contains exactly one zero, counted with multiplicity, where $\eta_n = \exp(-E_n/2)$ and E_n is chosen in Lemma 3.1 with $L = 2$.*

Proof. Put $H_j = T_j + T_{j+1}$. At $z = z_{j,\ell}(1+u)$,

$$\frac{H_j(z)}{T_j(z)} = 1 - (1+u)^{q_j}.$$

On $|u| = \eta_n/q_j$, one has $|q_j \log |1+u|| \leq 2\eta_n \leq 2$. The binomial theorem yields

$$|1 - (1+u)^{q_j}| \geq 1 + 2\eta_n - e^{\eta_n} \geq \eta_n/2 \quad (3.2)$$

eventually. The last inequality follows from $e^t - 1 - t \leq t^2$ for $0 \leq t \leq 1/4$, directly from its series. The right side of (3.1) is $O(e^{-E_n}) < \eta_n/2$. Thus $|F_p^{(n)} - H_j| < |H_j|$ on the entire boundary.

The model has the zero 0 of multiplicity m_j , if $m_j > 0$, and exactly the q_j simple nonzero zeros (1.8). The disk excludes 0. Distinct nonzero model zeros have distance at least $2\rho_j \sin(\pi/q_j) \geq 4\rho_j/q_j$ for $q_j \geq 2$, so it contains exactly one. Rouché's theorem proves the assertion. $\square$

Proposition 3.3 (Exclusion of additional zeros). *For every fixed annulus, the disks in Theorem 1.2, with a sufficiently small fixed choice of c , are disjoint and contain all its zeros for large n .*

Proof. The same-circle separation was proved above. Between circles, (2.9) gives a separation of order $Q/\sqrt{n}$, whereas disk radii are $O(\eta_n/Q)$. Their ratio tends to zero, proving disjointness.

Fix $r \in [a, b]$ and choose a largest term $|T_k|$ on $|z| = r$. Such a term exists since the summands tend to zero. The normalized block slopes

$$\frac{1}{q_j} \log \frac{|T_{j+1}|}{|T_j|} = \log r - \log \rho_j$$

strictly decrease. They show that the moduli increase up to a maximum and then decrease. The first crossing tends to zero. From (2.10), the maximizing index and its fixed number of neighbors satisfy $m_k \asymp \sqrt{n}$ and $q_k \asymp Q$, uniformly in r .

Write $u = |T_{k-1}|/|T_k| \leq 1$ and $v = |T_{k+1}|/|T_k| \leq 1$. The crossing separation gives

$$\frac{\log u}{q_{k-1}} + \frac{\log v}{q_k} = \log \rho_{k-1} - \log \rho_k \leq -cQ/\sqrt{n}.$$

Both u and v cannot be at least $1/4$, since that would bound the left side below by $-C/Q$, while $Q^2/\sqrt{n} \rightarrow \infty$.

All terms beyond these two neighbors have total modulus $O(e^{-c'Q^2/\sqrt{n}})|T_k|$. To verify this last statement, on the block immediately preceding $k-1$ its normalized slope exceeds that of block $k-1$ by at least $cQ/\sqrt{n}$, by (2.11). It is therefore positive by this amount. Earlier normalized slopes are larger still. Multiply by the global lower gap bound and sum a geometric series. The right side is identical with reversed signs. This argument uses local curvature only at the two initial blocks and global monotonicity thereafter.

If $u, v < 1/4$, the largest term exceeds the sum of all others, so there is no zero. Otherwise select the neighboring pair $j, j+1$ containing a neighbor of relative size at least $1/4$. It has $|q_j \log(r/\rho_j)| \leq \log 4 < 2$, so Lemma 3.1 bounds its remainder by $R_n|T_j|$, with $R_n = O(e^{-E_n})$ after choosing a common smaller $E_n = cQ^2/\sqrt{n}$.

At a zero choose the nearest model angle and write $z = z_{j,\ell}e^{v_0/q_j}$, where $|\operatorname{Re} v_0| \leq \log 4$ and $|\Im v_0| \leq \pi$. For $v_0 = x + iy$ in this rectangle,

$$|1 - e^{v_0}|^2 = (e^x - 1)^2 + 4e^x \sin^2(y/2) \geq \frac{1}{16}|v_0|^2.$$

Here $|e^x - 1| \geq |x|/4$ and $|\sin(y/2)| \geq |y|/\pi$ suffice. At the zero, $|1 - e^{v_0}| \leq R_n$, hence $|v_0| \leq 4R_n$ and $|z - z_{j,\ell}| \leq 8\rho_j R_n/q_j < \rho_j \eta_n/q_j$ eventually. Also $\rho_j = r \exp(O(Q^{-1})) \in [a/2, 2b]$. This proves the exclusion assertion. $\square$

4. COVERING, COUNTING AND THE ORIGIN

Let $|w| = r \leq R$. If $n + r\sqrt{n} \geq \nu_{j_0}$, choose j with $\nu_j \leq n + r\sqrt{n} < \nu_{j+1}$; otherwise take $j = j_0$. Lemmas 2.1 and 2.4 give

$$|\rho_j - r| \leq CQ/\sqrt{n}, \quad \rho_j \leq R + o(1), \quad q_j \asymp Q.$$

For $w \neq 0$, a nearest model angle has angular error at most π/q_j . Proposition 3.2 therefore supplies a true zero within $C(Q/\sqrt{n} + Q^{-1})$ of w . For $w = 0$, use any center of the first pair; its radius is $O(Q/\sqrt{n})$. This proves (1.4) on the whole disk, without extending an annular estimate across the origin.

For the lower bound in (1.5), choose two consecutive crossing radii in a fixed smaller annulus inside $A_{a,b}$. Such radii exist for all large n by (2.8). Their separation is at least $cQ/\sqrt{n}$. A point at the midpoint radius is at least half this radial separation from all model circles. Proposition 3.3 and the exponentially small disk radii preserve a lower bound $c'Q/\sqrt{n}$ for the distance to true zeros. Zeros outside a fixed enlargement of the annulus are a fixed positive distance away. This proves the sharp scale.

To count, let $J_n(a, b) = \{j : a < \rho_j < b\}$. It is an interval of consecutive indices, and (2.8) implies

$$\sum_{j \in J_n(a, b)} q_j = (b - a)\sqrt{n} + O(Q), \quad \#J_n(a, b) = O(\sqrt{n}/Q). \quad (4.1)$$

For the first assertion, the sum telescopes to the difference of its endpoint support exponents; the first and last m_j differ from $a\sqrt{n}$ and $b\sqrt{n}$ by $O(Q)$, using the same statement for their adjacent radii. Each circle contributes $\omega q_j/(2\pi) + O(1)$ model centers to an arc of length ω . Equations (4.1) give (1.10) for centers.

Passing to true zeros changes this count by at most $O(Q + \sqrt{n}/Q)$. At either radial boundary at most one model circle is within the maximal disk radius for large n , by (2.9); each carries $O(Q)$ centers. At either angular boundary each relevant circle has only $O(1)$ affected centers, since the angular displacement is $O(\eta_n/Q)$. This also handles centers or zeros lying exactly on a boundary, and proves (1.10). Approximation of a continuous compactly supported function by step functions on finitely many annular sectors proves (1.11); (1.10) on a containing annulus uniformly bounds the masses needed for this approximation.

At the origin the exact multiplicity is

$$\text{mult}_0 F_p^{(n)} = \min_{\nu_j \geq n} (\nu_j - n) = m_{j_0}, \quad (4.2)$$

where multiplicity zero means a nonzero value. In particular, taking $n = \nu_j + 1$ gives multiplicity $q_j - 1 \rightarrow \infty$. Thus a claim of eventual simplicity on the whole plane would be false. This does not affect the disk covering theorem or the annular statements.

5. THE CRITICAL GAP EXPONENT

We now put $p = 4/3$, so $\beta = 1/4$. The two-term remainder need not tend to zero on a fixed annulus. Instead, a rescaled derivative has subsequential limits among bilateral Gaussian series with arbitrary phases.

Lemma 5.1 (Bilateral Gaussian series have zeros). *Let $\kappa > 0$, $t \in \mathbb{R}$, and $|c_s| = 1$ for every $s \in \mathbb{Z}$. Then*

$$G(v) = \sum_{s \in \mathbb{Z}} c_s \exp(ts - \kappa s^2/2) e^{sv} \quad (5.1)$$

is a nonzero entire function and has a zero in $\mathbb{C}$.

Proof. Gaussian decay proves local uniform convergence and a bound $|G(v)| \leq \exp(C(1 + |v|^2))$. The function is $2\pi i$ -periodic. Its Fourier coefficients on the imaginary axis are precisely the nonzero numbers $c_s \exp(ts - \kappa s^2/2)$; absolute convergence justifies integration, and in particular $G \not\equiv 0$.

If G had no zeros, it would admit an entire logarithm h , with $\operatorname{Re} h(v) \leq C(1+|v|^2)$. We recall a short proof that h is quadratic. Write $h(v) = \sum a_k v^k$, $u = \operatorname{Re} h$, and $M_R = \max_{|v|=R} u(v)$. For $k \geq 1$, the Fourier formula and the mean-value property give

$$|a_k|R^k = \left| \frac{1}{\pi} \int_0^{2\pi} u(Re^{i\theta}) e^{-ik\theta} d\theta \right| \leq 2(M_R - u(0)) = O(1 + R^2).$$

Thus $a_k = 0$ for $k \geq 3$. Periodicity implies $h(v + 2\pi i) - h(v) \in 2\pi i\mathbb{Z}$, so this polynomial difference is constant. The quadratic coefficient vanishes and the linear coefficient is an integer. Consequently $G(v) = C_0 e^{kv}$ for some $k \in \mathbb{Z}$, contradicting the nonzero Fourier coefficient at every integer. Hence G has a zero. $\square$

Lemma 5.2 (Compactness of critical rescalings). *Let $n \rightarrow \infty$ along any sequence and choose w_n in a fixed annulus $A_{a,b}$. Put $r_n = |w_n|$ and choose $j = j(n)$ with*

$$\nu_j \leq n + r_n \sqrt{n} < \nu_{j+1}, \quad q = q_j. \quad (5.2)$$

Then $\nu_j \geq n$ eventually. The entire functions

$$G_n(v) = \frac{F_{4/3}^{(n)}(w_n e^{v/q})}{T_j(w_n e^{v/q})} \quad (5.3)$$

have a subsequence converging locally uniformly to a function (5.1) with $c_0 = 1$, $\kappa = (4/3)^2/r$ for some $r \in [a, b]$, and finite real t .

Proof. By Lemma 2.1, $q \sim (4/3)n^{1/4}$ and $m_j = r_n \sqrt{n} + O(q)$. For each fixed integer s , Taylor's theorem for $x^{4/3}$ and the bounded floor errors give

$$\Delta_{n,s} := \nu_{j+s} - \nu_j = sq + O_s(1), \quad \Delta_{n,s}/q \rightarrow s. \quad (5.4)$$

Every fixed s is active eventually: there are $\asymp \sqrt{n}/q \rightarrow \infty$ active support steps to the left of j .

Set $t_n = q d_{n,r_n}(\nu_j)$. Formula (2.5), together with $0 \leq n + r_n \sqrt{n} - \nu_j < q$, shows $t_n = O_{a,b}(1)$. After passage to a subsequence, $r_n \rightarrow r$ and $t_n \rightarrow t$. Uniformly on each fixed number of adjacent sparse blocks, (2.6) is $(1 + o(1))/(r_n \sqrt{n})$. Summing the full differences, for positive and negative integer increments alike, gives

$$\Phi_{n,r_n}(\nu_{j+s}) - \Phi_{n,r_n}(\nu_j) = st_n - \frac{q^2}{2r_n \sqrt{n}} s^2 + o(1) \rightarrow st - \kappa s^2/2. \quad (5.5)$$

For clarity, the discrete quadratic term before taking limits is $-\Delta_{n,s}(\Delta_{n,s} - 1)/(2r_n \sqrt{n})$; replacing it by the term displayed above changes it by $o(1)$.

The coefficient multiplying $\exp(\Delta_{n,s}v/q)$ in (5.3) has modulus given by (5.5) and phase $\exp(i\Delta_{n,s} \arg w_n)$. A diagonal subsequence makes these unit phases converge for every integer s , giving coefficients c_s with $c_0 = 1$. We do not replace the integer parts in these phases by real powers.

We verify the tails needed to pass from coefficient convergence to local uniform convergence. Fix $A > 0$ and $|\operatorname{Re} v| \leq A$. The real part of each exponent is the logarithmic weight at radius $r_n \exp(\operatorname{Re} v/q)$, normalized by the j th term. For a fixed sufficiently large integer S , (5.5) gives, at $s = S$ and $s = -S$, an upper bound $C_A S - cS^2$, with $c > 0$ depending only on the annulus. The threshold in n may depend on S, A . Formula (2.5) and local curvature likewise show that beyond the right endpoint the full slope is at most $-c'S/q$, and before the left endpoint it is at least $c'S/q$, after choosing S large compared with A . Global monotonicity preserves these bounds all the way to the ends of the active support. Every active gap is at least a fixed positive multiple of q . Each further sparse step therefore loses at least $c''S$. Consequently

$$\limsup_{n \rightarrow \infty} \sup_{|\operatorname{Re} v| \leq A} \sum_{|s| \geq S} |\text{the } s\text{th normalized term}| \leq \frac{C \exp(C_A S - cS^2)}{1 - \exp(-c''S)}. \quad (5.6)$$

Here inactive left terms are omitted. The right side tends to zero with S . Finite-term convergence and this tail bound prove local uniform convergence to (5.1). $\square$

Proof of the critical upper bound. Suppose the upper bound in (1.6) fails. There are $n_h \rightarrow \infty$ and $w_h \in A_{a,b}$ with $\text{dist}(w_h, \mathcal{Z}(F_{4/3}^{(n_h)})) > hn_h^{-1/4}$. Apply Lemma 5.2 and then Lemma 5.1. The nonzero limit G has a zero. Choose a small circle around that zero on which G has no zeros. Local uniform convergence and Rouché's theorem give zeros v_h of G_{n_h} in a fixed bounded disk. Equation (5.3) yields zeros $\zeta_h = w_h \exp(v_h/q_{j(n_h)})$ of the original derivatives, satisfying $|\zeta_h - w_h| \leq C/q_{j(n_h)} \leq C'n_h^{-1/4}$. This contradicts the choice of w_h . $\square$

6. A ZERO-COUNT UPPER BOUND AND FINAL DEDUCTIONS

Lemma 6.1. *For every fixed $R > 0$ and $4/3 \leq p < 2$, the number of zeros of $F_p^{(n)}$ in $|z| \leq R$, counted with multiplicity, is $O_{R,p}(\sqrt{n})$.*

Proof. For $n \geq 1$ and $r > 0$, Cauchy–Schwarz and the negative-binomial series give

$$\begin{aligned} S_n(r) &:= \sum_{m \geq 0} \frac{\sqrt{(n+m)!}}{m!} r^m \\ &\leq \sqrt{n!} (1-t)^{-(n+1)/2} \exp(r^2/(2t)), \quad 0 < t < 1. \end{aligned} \quad (6.1)$$

Indeed, after dividing by $\sqrt{n!}$, each term is $\sqrt{\binom{n+m}{n}}/m! r^m$, and the two squared sums are $\sum_m \binom{n+m}{n} t^m = (1-t)^{-n-1}$ and $\sum_m (r^2/t)^m/m! = \exp(r^2/t)$. Taking $t = r/(r + \sqrt{n})$ and using $\log(1+x) \leq x$ yields

$$\log S_n(r) \leq \frac{1}{2} \log(n!) + r\sqrt{n} + (r + r^2)/2. \quad (6.2)$$

This estimate also appears in Hou [4, Lemma 2.1]; its elementary derivation is included here.

Factor $F_p^{(n)}(z) = z^{m_0} g_n(z)$ with $m_0 = m_{j_0} = O(Q) = o(\sqrt{n})$. For large n , $m_0 \leq \sqrt{n}$ and

$$g_n(0) = \frac{\sqrt{(n+m_0)!}}{m_0!} \geq \sqrt{n!},$$

since $m_0! \leq m_0^{m_0} \leq n^{m_0/2}$, with the case $m_0 = 0$ read separately. Choose a fixed $L > \max(1, R)$, for instance $L = 2(R + 1)$. On $|z| = L$, $|g_n(z)| \leq L^{-m_0} S_n(L) \leq S_n(L)$. Jensen's formula bounds the number of zeros of g_n in $|z| \leq R$ by

$$\frac{\log S_n(L) - \log g_n(0)}{\log(L/R)} = O_R(\sqrt{n}).$$

Adding the origin multiplicity proves the lemma. $\square$

The lower bound in (1.6) follows from area. The upper bound already proved implies $h_n(A_{a,b}) < 1$ eventually. Zeros in the disk $|z| \leq b + 1$ suffice to cover $A_{a,b}$ by disks of radius $h_n(A_{a,b})$. There are at most $C\sqrt{n}$ such zeros by Lemma 6.1. Therefore $\pi(b^2 - a^2) \leq C\sqrt{n} \pi h_n(A_{a,b})^2$, giving the asserted lower bound.

Finally, every nonempty open set contains a nonzero point w and a disk $D(w, \varepsilon)$ contained in a fixed annulus. The upper bounds in Theorem 1.1 tend to zero for every sufficiently large integer n . They place a derivative zero in that disk, proving (1.1) in both regimes. The subsequence in the critical proof is used to contradict failure of a uniform bound; the final assertion is not restricted to a subsequence.

Further questions. For $p = 3/2$, the sharp annular covering scale is $n^{-1/6}$; at $p = 4/3$ it is $n^{-1/4}$. The latter proof supplies neither an explicit circular grid nor eventual simplicity or a limiting zero measure at the endpoint. These conclusions are not asserted. The range $1 < p < 4/3$ requires another analysis. A quantitative, effective version of Lemma 5.1 would also give explicit constants in the critical covering theorem.

Source and assistance statement. An earlier supplied note contains the exponential construction $\sum_{k \geq 1} \exp(-10k^2 + k^{3/2}z)$; a second supplied note proposes the sparse construction for $p > 4/3$. The present arguments develop that sparse construction and its endpoint. Language-model assistance was used in the derivations and manuscript preparation. No external peer review or machine verification of this manuscript is asserted.

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